TD 2: Software Processes — Solutions

C-1: Anatomy of a Process

(a) Describe two activities

Activity	Role	Input artifacts	Output artifacts	Precondition	Postcondition
Elicit and document requirements	Business analyst	Client needs	Requirements specification	Client is available	Requirements are documented and validated
Load-test synchronization service	Tester	Service and test plan	Test report	Service is ready for testing	Performance results are recorded

(b) Framework or umbrella?

Activity	Answer
Implement the list-sharing endpoint	F
Review a teammate's pull request	F
Protect the main branch	U
Review the risk register every week	U
Deploy the app on production servers	F

Track project progress against the plan	U
Write tests that check a requirement	F
Measure time spent per task	U

F = Framework, U = Umbrella.

C-2: From a Feature to Tasks

(a) Who does what?

Task	Role
Clarify rules with the client	Business analyst
Design reminder and quiet-hours screens	UX/UI designer
Implement reminder scheduling	Developer
Test quiet hours and time zones	Tester
Ensure delivery on schedule	Project manager

(b) Problems with the task board

1. Sara has too many tasks: Redistribute tasks among team members.
2. Backend task has taken two weeks: Identify blockers and split it into smaller tasks.
3. “Fix bug” is vague: Describe the bug and expected result.
4. Bug has no owner: Assign a responsible team member.

5. Tests are done without review: Require a review before closing the task.
6. Time-zone question is still in backlog: Contact the client quickly because this affects the reminder logic.

C-3: Which Process Model?

System	Model	Justification
1. Environmental travel-planning app	Incremental	Features can be delivered progressively and improved with user feedback.
2. University accounting system with unchanged rules	Waterfall	Requirements are stable and well understood.
3. Car ABS system	V-model	It is safety-critical and requires rigorous testing.
4. Hospital appointment app with uncertain needs	Prototyping	Prototypes help discover what patients and doctors need.
5. National exam-results portal	Spiral	It helps identify and reduce major performance and availability risks.

Iterative or incremental?

- Iterative: Improve an existing feature, such as making synchronization faster.
- Incremental: Add new features, such as shared lists and then subscription payments.

C-4: The First Turn of the Spiral

(a) Identify and score risks

Risk exposure = Probability × Impact.

Risk	Category	Probability	Impact	Score
Translation delay is too high	Technical	4	5	20
Darja translation is inaccurate	Technical	4	5	20
Translation services are too expensive	Cost	3	4	12
Project misses the deadline	Schedule	3	4	12
Private conversations are exposed	Security	3	5	15

(b) Reduce the two highest risks

Risk 1: Translation delay

Build a prototype connecting two laptops to a translation service. Measure the delay over 50 sentences. If the median exceeds 2 seconds or the 95th percentile exceeds 4 seconds, try another provider or use subtitles instead of translated speech.

Risk 2: Inaccurate Darja translation

Test a small set of Darja recordings and compare the results with translations checked by fluent speakers. If accuracy is too low, improve the data, try another provider or limit the first version to better-supported languages.